

Faculty of Information Technology & Computer Science

Software Engineering Department

SESE3133 – Formal Methods in SE

Assignment – 4

Submitted By:

ID: L1_____BSSE_____ Name: _____

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Section: _____ Total Marks: 50 Obtained Marks: _____

CLO#	Course Learning Outcome (CLO)	Taxonomy Level
CLO 3	Demonstrate how formal languages can be used specify the systems and verify correctness properties.	C3

Submission deadline:

11:59 pm, Sunday, June 21, 2026

Submission Instructions:

- Print this page and attach as front page of your assignment.
- This assignment can be done in a group of 2.
- All Solutions should be handwritten.
- Only use A4 white papers.
- Write your name, ID & Section on each page.
- Combine all solutions into a single PDF file using the the CamScanner App.
- Only one group member will upload the PDF file on Portal.
- Submit the hard copy on next day as well.

Q#1: Automated Smart Medicine Dispenser [CLO 3 – 20 Points]

Consider an automated hospital medicine dispenser comprising three components: a Biometric Scanner (BS), a Central Controller (CC), and a Physical Dispenser (PD). Each component has two states (`0` and `1`), with `0` being the initial state.

- Biometric Scanner (BS): Moves from `0` to `1` via an independent action `scan_finger`. It returns to `0` via a shared handshake action `verify_auth`.
- Central Controller (CC): Waits at `0`. When `verify_auth` occurs, it transitions to `1`. From state `1`, it issues a shared handshake action `release_cmd` to return to `0`.
- Physical Dispenser (PD): Waits at `0`. When `release_cmd` occurs, it transitions to `1`. It then drops the medicine via an independent action `dispense` and returns to `0`.

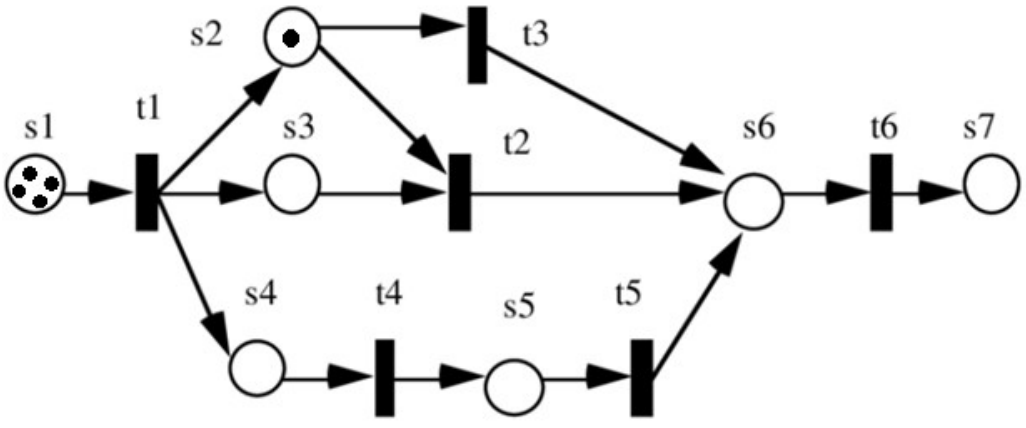
Task:

1. **Model** the individual Transition Systems for *BS, CC,* and *PD*. [9 Points]
2. **Construct** the reachable state space of the joint Transition System using Handshake Composition: *BS* ‖ *CC* ‖ *PD* [11 Points]

   Represent each composite state as a 3-tuple (S_{BS},S_{CC},S_{PD}), starting from the initial state (*0,0,0*). *(Note: Refer to Week 13, Slide 38 for help)*

Q#2: Concurrent Process Synchronization Pipeline [CLO 3 – 15 Points]

Consider the Petri Net shown in the figure, which models a multi-stage concurrent processing pipeline. The system tracks token distribution across seven distinct places (S_1 through S_7) and executes state transitions via six processing units (t_1 through t_6). All arcs carry a standard weight of 1.



Initial System State (Step 0): The pipeline is initialized with four raw data items waiting in the input buffer, and one resource ready in the validation layer. This translates to the initial marking M_0): $S_1 = 4$ tokens, $S_2 = 1$ token. All remaining places contain S_3, S_4, S_5, S_6, S_7 contain 0 tokens.

Execution Constraint: Transitions must execute using discrete, maximum-parallel execution steps. In scenarios where a token conflict arises at S_2 between t_2 and t_3, prioritize resolving the joint execution path t_2 whenever its synchronization partner token at S_3 is present.

Task:

1. **Apply** the enabling rules of Petri Nets to determine the exact set of transitions capable of firing at Step 0. [3 Points]
2. **Compute** the token distribution and state changes by constructing a complete firing table that tracks all places (S_1 to S_7) through Step 1, Step 2, and Step 3. [12 Points]

Q#3: Z Specification for a Staff Directory System [CLO 3 – 15 Points]

A company requires a formal, unambiguous specification for its internal corporate directory system. The system manages staff contact details and must enforce strict integrity constraints to prevent data duplication and configuration errors.

Let *NAME* represent the universal set of all staff names, and let *PHONE* represent the universal set of all valid telephone numbers.

System Requirements:

System State: The system tracks the complete set of registered staff members and maps each individual to exactly one unique phone number. A staff member can only exist in the registry if they have a phone number mapped to them.

State Transition (Addition): The system must allow an administrator to input a new staff member's name and their corresponding phone number. This action can only proceed if the staff member is not already registered in the directory. Upon successful completion, the system state updates to include the new record.

Task:

1. **Formulate** the formal state space schema named `PhoneBook` that structurally defines the state variables, data types, and the structural invariant required to keep the staff registry synchronized. [5 Points]
2. **Construct** the operation schema named `AddName` that models the addition of a new staff member to the directory. Your schema must clearly specify all necessary state changes, inputs, preconditions, and postconditions using valid Z mathematical notation. [10 Points]